

Lesson 5 Quick Reference — Input & Operators

Syntax only. For the *why*, see the Lesson 5 reference sheet.

input()

```
name = input("Name: ")      # ALWAYS returns a str
age = int(input("Age: "))    # convert as you collect
price = float(input("$"))    # prices need float
```

JavaScript	Python
<code>prompt("Name: ")</code>	<code>input("Name: ")</code>
<code>parseInt(prompt(...))</code>	<code>int(input(...))</code>
<code>parseFloat(prompt(...))</code>	<code>float(input(...))</code>

It gives text even when they type digits.

`age + 1` → `TypeError`

`n * 2` on `"7"` → `"77"` — **no error**, just wrong

A **count** needs `int()`. A **price** needs `float()`.

`int("4.50")` → `ValueError`

The seven operators

```
3 + 4      # 7
3 - 4      # -1
3 * 4      # 12
2 ** 5     # 32    exponent
17 / 5     # 3.4   ALWAYS a float
17 // 5    # 3     whole part only   (no JS equivalent)
17 % 5     # 2     leftover
```

// and % answer two halves of one question:
17 // 5 each, 17 % 5 left over.

-7 % 3 → 2 in Python, -1 in JavaScript.

Precedence

() → ** → * / // % → + -

```
9 + 6 / 3 * 2 - 1      # 12.0
```

-2 ** 2 is -4 . Write (-2) ** 2 for 4 .

Error → cause

Error	It means
<code>TypeError: can only concatenate str</code>	math on <code>input()</code> without converting
<code>ValueError: invalid literal for int()</code>	<code>int()</code> on <code>"4.50"</code> — use <code>float()</code>
got <code>"77"</code> not <code>14</code>	forgot to convert <code>input()</code>
got <code>4.8</code> slices	used <code>/</code> where <code>//</code> belongs
<code>ZeroDivisionError</code>	divided by zero

Before you push

name, assignment, date · every `input()` number converted · counts use `//` · money uses `:.2f` ·
it **runs** · **pushed**